HealthConnect Rwanda

Final KPI Summary & Executive Summary

Project:	Improving Patient Appointment Attendance & Healthcare Support	**Track:**	Data Analytics
Program:	AnalystLab Africa Experience Lab	**Review:**	Week 8 – Final Integration
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1. EXECUTIVE SUMMARY

The HealthConnect Rwanda Data Analytics project focuses on improving healthcare operations and patient support through data analysis and business intelligence. The project uses healthcare data to examine patient demographics, appointment activity, clinical records, financial performance, operational indicators, and patient experience.

A six-page Power BI executive dashboard was developed to support data-driven review across these areas. The dashboard brings together key indicators and interactive views for executive and operational users.

The project identified several important areas requiring management attention. Financial data shows a gap between gross billed amounts and payments collected, while incomplete appointment-to-billing relationships limit reliable department-level revenue allocation. The available appointment data also describes visit types but does not contain a confirmed attendance-status field. As a result, attendance and no-show rates cannot currently be validated.

The final analytics package emphasizes validated findings, transparent data limitations, and practical recommendations. The dashboard can support decision-making, but some operational and financial conclusions require improved source data and stronger record linkage.

2. FINAL KPI SUMMARY

KPI / Indicator	Validated Result	Interpretation
Patient profiles	**20,000**	Patient demographic analysis population
Appointment records	**5,000**	Appointment dataset used for analysis
Male patients	**10,090 (50.45%)**	Slightly more male than female profiles
Female patients	**9,910 (49.55%)**	Patient profile distribution is nearly balanced
Average patient age	**Approx. 45 years**	Descriptive demographic indicator
Emergency visit type	**1,650 (33.00%)**	Share of appointment records categorized as Emergency
Follow-up visit type	**1,694 (33.88%)**	Largest of the three visit-type categories
Consultation visit type	**1,656 (33.12%)**	Share categorized as Consultation
Gross billed amount	**Approx. RWF 9.31B**	Total billing amount reported
Payments collected	**Approx. RWF 8.39B**	Total recorded payments
Collection rate	**Approx. 90.1%**	Collected amount relative to gross billed amount
Outstanding amount	**Approx. RWF 919.95M**	Reported unpaid balance

KPI / Indicator	Validated Result	Interpretation
Dept. Allocated Revenue	**Approx. RWF 771.13M**	Revenue associated with matched appointment records
Unallocated billing revenue	**Approx. RWF 8.54B**	Billing amount not reliably allocated to departments
Insurance approval rate	**49.98%**	Approval indicator; pending/blank statuses require care
Diagnosis records	**Approx. 60,000**	Diagnosis entries, not unique patients/prevalence
Orphaned diagnosis categories	**0**	Diagnosis category relationships validated

Important: The values above are descriptive results from the available datasets. They do not establish causation or prove improvements in attendance, patient outcomes, or revenue collection.

3. KEY BUSINESS INSIGHTS

3.1 Financial Performance and Data Reconciliation

The reported gross billed amount is approximately RWF 9.31 billion, while recorded payments total approximately RWF 8.39 billion. The reported outstanding amount is approximately RWF 919.95 million. However, only approximately RWF 771.13 million is allocated to matched appointment records. The large unallocated amount indicates that current appointment-to-billing relationships are not sufficient for dependable department-level revenue analysis.

> **Business Implication:** Management should prioritize financial data reconciliation and record linkage before using department-level revenue figures for performance comparisons.

3.2 Appointment Mix

Emergency, Follow-up, and Consultation visits account for approximately 33% each of the 5,000 appointment records. Follow-up visits represent the largest category at 33.88%.

> **Business Implication:** The appointment mix can inform service planning, but it cannot show which visit type experiences the highest no-show rate because attendance outcomes are not available.

3.3 Patient Demographics

The patient profile dataset contains 20,000 records, with a nearly balanced gender distribution. The average patient age is approximately 45 years.

> **Business Implication:** These demographic indicators provide a baseline for understanding the population represented in the dataset. They should not be used alone to make assumptions about service needs or patient behavior.

3.4 Clinical Records

The diagnosis dataset contains approximately 60,000 diagnosis records across the reported categories. The category counts are relatively similar.

> **Business Implication:** These figures describe recorded diagnosis entries. They should not be interpreted as unique patient counts, disease prevalence, or clinical risk without additional validation.

3.5 Insurance Claims

The reported insurance approval rate is 49.98%. Pending and blank claim statuses require careful interpretation and should not automatically be classified as rejected claims.

> **Business Implication:** Improving claim-status completeness and consistency would strengthen insurance reporting and follow-up.

4. FINAL RECOMMENDATIONS

1. **Capture appointment outcomes consistently:** Introduce or standardize appointment statuses such as Attended, No-Show, Cancelled, and Rescheduled. This would enable reliable attendance and no-show KPIs.
2. **Improve appointment reminders and administrative follow-up:** Once reliable appointment status and contact data are available, evaluate reminder and follow-up workflows. Their impact should be measured rather than assumed.
3. **Reconcile billing and appointment records:** Investigate unmatched AppointmentIDs and establish reliable relationships between billing, payments, appointments, and departments.
4. **Strengthen financial reporting controls:** Validate gross billing, payments, outstanding balances, and department-allocated revenue against agreed business definitions and source records.
5. **Improve insurance claim-status completeness:** Define consistent claim statuses and ensure blank or pending values are handled transparently in reporting.
6. **Improve operational resource data:** Capture bed occupancy, room usage, equipment utilization, maintenance, and downtime information before reporting actual utilization rates.
7. **Maintain clear dashboard documentation:** Document KPI definitions, data sources, validation outcomes, limitations, and the interpretation of each dashboard page.

5. DATA LIMITATIONS

- Appointment attendance status is not available in the confirmed appointment data; attendance and no-show rates cannot be validated.
- A substantial number of billing records do not match the available appointment records.
- Department-allocated revenue represents matched records only and is not equivalent to total gross billing.
- Facility mapping is limited and should not be used for unsupported facility rankings.
- Diagnosis counts represent records, not unique patients or prevalence.
- Resource availability fields do not provide enough information to calculate actual bed, room, or equipment utilization.
- Pending or blank insurance claim statuses should not automatically be treated as rejected claims.

6. CONCLUSION

The HealthConnect Rwanda analytics project provides a consolidated Power BI dashboard and a documented set of validated indicators to support healthcare management review.

The strongest immediate opportunities are to improve appointment outcome capture, reconcile financial records, strengthen insurance data quality, and expand operational resource tracking. Addressing these gaps would enable more reliable attendance monitoring, financial analysis, and operational decision support.

The dashboard and accompanying documentation provide a foundation for continued integration and future improvement. Any claims of measurable business impact should be made only after implementation and evaluation using reliable data.